

PORT VILA EFATE ISLAND, Vanuatu

WMO: 915570

Lat: 17.699S

Lon: 168.320E

Elev: 70

StdP: 14.66

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	58.6	60.5	55.6	65.9	65.5	57.3	70.2	65.7	20.5	76.1	18.7	76.2	1.5	60	0.285

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.6	89.1	79.5	88.0	79.1	87.1	78.6	81.3	86.4	80.6	85.9	80.0	85.1	8.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.0	156.1	84.4	79.1	151.4	83.8	78.6	148.7	83.4	45.1	86.6	44.3	85.8	43.7	85.7	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.2	17.3	15.6	54.9	90.9	2.4	1.3	53.2	91.9	51.8	92.6	50.4	93.3	48.7	94.3		
(5)			54.7	83.2	2.6	1.5	52.9	84.2	51.4	85.1	49.9	85.9	48.1	86.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	76.2	79.9	80.4	80.1	78.4	75.5	73.5	71.8	71.8	73.2	75.0	76.7	78.7
	DBStd	4.02	1.96	1.89	1.62	2.04	2.61	2.77	3.05	3.39	3.23	2.85	2.53	2.31
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9574	927	853	932	851	791	706	676	677	696	774	802	889
	CDD65	4099	462	433	467	401	326	256	211	212	246	309	352	424
	CDH74	31269	4481	4250	4361	3207	2069	1164	801	904	1356	2081	2724	3871
	CDH80	7503	1406	1366	1274	733	277	88	57	65	161	351	592	1133
Wind	WSAvg	5.8	5.5	5.1	5.0	5.0	5.4	5.4	5.4	6.2	6.2	7.1	6.8	6.0
Precipitation	PrecAvg	88.5	13.0	11.5	15.3	8.7	6.3	6.0	3.6	4.5	4.2	4.8	6.1	7.0
	PrecMax	131.7	24.9	31.4	32.7	19.4	18.1	16.9	8.8	13.2	13.2	22.3	20.1	22.3
	PrecMin	43.0	2.3	2.9	4.5	2.4	1.9	0.8	0.3	0.5	0.5	0.6	0.6	1.0
	PrecStd	20.0	6.3	6.0	6.9	5.1	4.1	3.8	2.5	3.3	3.1	4.5	4.6	5.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.0	91.5	89.7	88.4	86.0	84.5	84.0	84.2	86.0	87.1	87.9	89.5
		MCWB	79.7	81.1	79.8	79.6	78.1	77.1	76.1	76.3	76.2	77.1	77.6	78.7
	2%	DB	88.3	89.1	88.1	86.7	84.3	82.4	81.6	81.9	83.6	84.7	86.2	87.8
		MCWB	79.1	80.1	79.6	78.9	77.0	75.6	74.7	74.4	75.1	75.1	76.8	78.1
	5%	DB	87.3	87.7	87.0	85.5	82.8	80.8	79.8	80.2	81.9	83.3	84.6	86.4
		MCWB	78.7	79.4	79.3	78.3	75.8	74.5	73.2	73.0	73.8	74.4	76.0	77.7
	10%	DB	86.1	86.4	85.8	84.1	81.5	79.2	78.2	78.6	80.3	82.0	83.5	85.5
		MCWB	78.1	78.8	78.8	77.7	75.0	73.5	71.9	71.8	72.6	73.6	75.3	77.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.8	82.8	81.9	81.0	80.1	78.5	78.1	77.4	78.0	78.8	79.9	80.9
		MCDB	87.4	88.3	86.2	85.1	82.8	82.3	82.0	82.2	82.9	84.7	85.5	86.6
	2%	WB	80.8	81.3	80.9	80.1	78.7	77.1	76.0	75.7	76.5	77.3	78.7	79.9
		MCDB	86.2	86.8	85.9	84.5	82.5	80.7	79.9	79.9	81.6	82.4	83.9	85.5
	5%	WB	80.0	80.5	80.1	79.3	77.6	76.1	74.7	74.6	75.4	76.3	77.7	79.2
		MCDB	85.2	86.0	85.2	83.7	81.2	79.4	78.3	78.3	80.0	81.0	82.6	84.5
	10%	WB	79.2	79.8	79.5	78.6	76.6	75.1	73.5	73.6	74.3	75.3	76.8	78.4
		MCDB	84.2	84.8	84.4	83.0	80.0	78.2	76.9	77.2	78.7	79.7	81.4	83.5
Mean Daily Temperature Range	5% DB	MDBR	12.9	12.6	11.8	11.9	12.5	11.8	12.6	13.1	14.0	14.1	13.6	13.6
		MCDBR	14.6	14.4	13.9	13.7	13.8	13.2	14.3	14.3	15.5	16.1	15.7	15.2
		MCWBR	6.9	6.9	6.8	7.2	7.5	7.6	8.1	7.7	8.1	7.8	7.8	7.2
	5% WB	MCDBR	12.7	12.9	12.5	12.1	11.6	11.1	11.8	11.3	12.8	12.6	13.0	12.9
		MCWBR	6.6	6.8	6.8	6.9	7.2	7.3	7.5	6.8	7.5	6.9	7.2	6.8
Clear-Sky Solar Irradiance	taub	0.402	0.394	0.389	0.386	0.389	0.393	0.383	0.380	0.388	0.395	0.396	0.390	
	taud	2.557	2.591	2.616	2.618	2.573	2.552	2.557	2.563	2.524	2.530	2.546	2.577	
	Ebn at Noon	299	298	293	282	270	262	269	280	288	295	299	303	
	Edh at Noon	35	33	31	29	29	28	29	30	34	35	35	34	
All-Sky Solar Radiation	RadAvg	2033	1957	1766	1505	1280	1143	1210	1416	1717	1882	2021	2100	
	RadStd	180	135	141	127	120	85	86	101	153	149	164	132	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	+0.59	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	+0.59	N/A	N/A	N/A	N/A

Nomenclature: See separate page